

# BEMELINGUE Wilfried

Île-de-France, France

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## AI & Data Engineering Student – Looking for a 6-month end-of-studies internship

### PROFILE

Final-year engineering student (M2) at EPITA, with a strong analytical background and a deep interest in artificial intelligence and data. Known for adaptability, intellectual curiosity, and rigor in problem-solving. Experience on complex projects in collaborative environments, with the ability to clearly communicate technical concepts. Available for a 6-month internship starting immediately.

### PROFESSIONAL EXPERIENCE

**Intern – Data Pipeline Optimization, COMUTITRES (Paris)** 09/2024 – 02/2025

- Migration and optimization of Python/SQL workflows to Airflow on AWS to improve processing reliability.
- Design of automatic retry and monitoring mechanisms using JSON configurations.
- Creation of KPI dashboards for Business Intelligence and improvement of CI/CD pipelines with GitLab.

**Software Developer Intern, TDA CERMAI (Paris)** 05/2022 – 07/2022 & 04/2023 – 07/2023

- Development of full-stack monitoring applications (JavaScript front-end, Python/ObjectScript back-end).
- Integration of REST APIs for real-time processing and display of operational data.
- Contribution to improving system stability and application performance.

### ACADEMIC AND PERSONAL PROJECTS

**Neuro-Graph Trader – Autonomous GNN-RL Agent** – Extension of a Differential Graph Transformer architecture (Stanford CS224W) into a multi-agent decision-making system combining dynamic GNNs, reinforcement learning (PPO) and financial NLP, with spatio-temporal attention, XAI modules and a complete modular training and evaluation pipeline. (*Python, PyTorch, PyG, FinRL*)

**LLM Society – Multi-Agent LLM Framework** – Simulation of social dynamics with local LLMs (Ollama: phi, gemma3): each agent has a configurable psychological state (YAML) and interacts through structured JSON actions. Evaluation of coherence and emergent behaviors across multiple scenarios.

**Stochastic Modelling and Simulation** – Implementation of numerical solutions to stochastic differential equations with dynamic drift and volatility estimation to model non-stationary systems. (*Python, NumPy*)

**Deep Learning Framework from Scratch** – Development of a lightweight neural network library implementing automatic differentiation, computational graphs and gradient-based optimization in pure Python and NumPy.

### EDUCATION

**EPITA – Engineering Student (M2), SCIA-G specialization (AI and Graphs)** 2023 – 2026

- Advanced courses in artificial intelligence, machine learning, distributed systems and data engineering.
- Applied projects in graph learning, optimization and scalable systems.
- Intensive training through programming, mathematics, cloud and engineering fundamentals bootcamps.

**ECE – Paris (B2)** 2022 – 2023

- Web development (PHP, JavaScript), cloud computing (AWS) and Java application development.

**EPSI – Rennes (B1)** 2021 – 2022

- Web application development (Python, PHP), SEO and Linux/Windows server administration.

**Université Saint Jean – Yaoundé** – Preparatory class 2020 – 2021

### TECHNICAL SKILLS

**Machine Learning & AI** : Deep Learning, GNN, Reinforcement Learning, XAI, Reco. Systems, LLM, NLP

**Data Engineering** : Python, SQL, Airflow, Spark, Hadoop, Data Pipelines

**Cloud & DevOps** : AWS, Docker, Kubernetes, GitLab CI/CD

**Software Development** : Python, Java, JavaScript/TypeScript, C/C++, REST API, React

**Databases** : MySQL, PostgreSQL, MongoDB, Neo4j

**Visualization** : Tableau, Power BI, Matplotlib, Seaborn

**Mathematical Foundations** : Linear Algebra, Probability Theory, Optimization, Stochastic Processes

**Tools** : Linux, Git, Jupyter, Bash, Hugging Face

### LANGUAGES

French : Native language

English : Fluent (B2)